

Berk Kaya

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PROFESSIONAL SUMMARY

AI-focused Software Engineer specializing in computer vision and generative AI. Graduated from Bilkent University with honors. 3 years of industry experience.

EDUCATION

Bilkent University — B.Sc. Computer Engineering (Honors)

2016 – 2020 | GPA: 3.89 / 4.00 — High Honors

LANGUAGES

- English: C1 — Advanced (TOEFL iBT 105)
- Japanese: A2 — Elementary

WORK EXPERIENCE

Machine Learning Engineer — Huawei R&D; Center — Istanbul (2022 – Present)

Developed on-device vision models (MobileNet variants) for smartphone cameras. Achieved 15% mAP improvement on low-light object detection benchmarks.

Junior AI Engineer — Asisguard (2020 – 2022)

Built real-time person re-identification pipeline using PyTorch and TensorRT. Deployed on Jetson Nano edge devices for security camera systems.

PROJECTS

- StyleGAN Turkish Art Generator:** Fine-tuned StyleGAN3 on 50K Turkish miniature paintings dataset. HuggingFace Space with 5K+ monthly users.
- Sign Language Translator:** Real-time Turkish sign language recognition using MediaPipe + LSTM. 94.2% accuracy. Open-sourced on GitHub (180 stars).
- Satellite Image Segmentation:** U-Net based segmentation model for agricultural land classification. Kaggle top 8%.

INTERNSHIPS

AI Research Intern — ASELSAN AI Lab (Summer 2019)

Implemented attention mechanisms for radar waveform classification.

CV Intern — Meteksan Defense (Winter 2019)

Object detection pipeline for UAV imagery using YOLOv4.

TECHNICAL SKILLS

Python, PyTorch, TensorFlow, OpenCV, TensorRT, ONNX, Docker, CUDA, FastAPI, NumPy, scikit-learn, HuggingFace

AI & MACHINE LEARNING

Deep AI expertise: Published 1 workshop paper at ECCV 2023. Kaggle Expert badge. Experience with GANs, Transformers, diffusion models, LLMs, and edge AI deployment. Active contributor to PyTorch Vision repository.